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COMPREHENSIVE STUDY GUIDE

Pharmacognosy Practical

Techniques, Schedules & MCQ Revision

PART ONE · Foundations, Techniques & Plant Structure

Crude drugs · the microscope · magnification
Stains & reagents · micrometry · clearing agents
Plant structure · stomata · crystals · trichomes

PART TWO · The Practical Schedules & MCQ Recall

Fibres & dressings · starches · gums · mineral matters
Crude-drug & vegetable-drug macroscopy · experiment log
Quick-recall flash facts for the MCQ test

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PART ONE — FOUNDATIONS, TECHNIQUES & PLANT STRUCTURE

- 1 Introduction to Practical Pharmacognosy**
Crude drugs · botanical vs chemical aspects · lab rules
- 2 Use & Care of the Microscope**
Parts & functions · magnification · focusing & the condenser
- 3 Practical Techniques**
Clearing agents · stains & reagents · maceration · micrometry
- 4 Diagnostic Structures of Plant Material**
Stem tissues · leaf & stomata · flowers · cell inclusions

PART TWO — THE PRACTICAL SCHEDULES & MCQ RECALL

- 5 Schedule I — Fibres & Surgical Dressings**
Quality standards · chemical tests · absorbency & elasticity
- 6 Schedule II — Study of Starches**
Amylose vs amylopectin · granule morphology · iodine test
- 7 Schedule III — Unorganised Crude Drugs (Gums)**
Acacia vs Tragacanth · identification tests
- 8 Schedule IV — Mineral Matters**
Kaolin, talc, chalk, bentonite · sedimentation · uses
- 9 Schedule V — Examination of Crude Drugs**
Beans · turmeric · hibiscus · fruits · leaves
- 10 Schedule VI — Macroscopy of Vegetable Drugs**
Diagnostic characters · Tithonia & Talinum
- 11 Student Experiment Results Summary**
Practicals 1–7 · aims, methods & conclusions
- 12 Quick-Recall Flash Facts**
Microscope · minerals · starch · gums · dressings for the MCQ

§1 INTRODUCTION TO PRACTICAL PHARMACOGNOSY

Practical pharmacognosy is where the raw materials of medicine are recognised, checked and understood before they ever reach a formulation. This opening section sets out what a crude drug is, the two complementary angles from which every specimen is studied, and the working rules of the laboratory.

■ 1.1 What crude drugs actually are

A **crude drug** is a natural substance of plant, animal or mineral origin that has had little done to it beyond drying or a simple preparatory step. These are the unrefined starting materials from which medicines and pharmaceutical products are built. A plant crude drug may be tied to one specific morphological part — SENOKOT tablets, for instance, are prepared from the pods of *Cassia senna* — or it may supply a purified chemical, as with CAFERGOT, which pairs caffeine from *Coffea arabica* with ergometrine from *Claviceps purpurea*.

■ 1.2 The two aspects of the subject

Aspect	What it covers
Botanical	Naming of the drug and its macro- and micro-morphological examination. The student learns to identify a plant material and describe the diagnostic features that authenticate it.
Chemical & physico-chemical	The chemical make-up of the crude drug — qualitative identification and quantitative estimation of the constituents that give it its activity.

■ 1.3 Pharmacopoeias

Pharmacopoeias — African, British, European, Indian and others — carry monographs on plant drugs that record their botanical and geographical source, chemical assay, means of identification, and pharmaceutical or medicinal use. A pharmacist is expected to read these monographs and run the stated pharmacopoeial tests before a drug material is accepted into a formulation.

■ 1.4 General laboratory rules

- » Coursework carries **40%** of the total mark for the course, so the practicals matter.
- » All results are to be entered into the data book **during** the three-hour practical, not afterwards.
- » A knee-length white laboratory coat must be worn at all times; no food or drink is allowed in the lab.
- » Equipment must be treated with care and every reagent bottle returned to its correct place after use.
- » Read the schedule before class. Nail varnish and eye cosmetics that could stain or scratch the lenses are banned.
- » Use only your assigned microscope and clean optical surfaces with proper lens tissue or cloth.

§2 USE & CARE OF THE MICROSCOPE

The microscope is the central instrument of the practical course, so knowing its parts, how magnification is built up, and how to focus and maintain it is foundational to everything that follows.

■ 2.1 Parts and their functions

Part	Function
Eyepiece / ocular lens (×10)	The lens viewed through at the top; it further enlarges the image formed by the objective.
Objective lenses (×4, ×10, ×40)	Gather light from the specimen, magnify it and form a real, enlarged image.

Part	Function
Nosepiece	Carries the objectives and is rotated to switch between magnifications.
Stage	The flat platform holding the slide, with a central aperture that lets light through.
Iris diaphragm	Regulates how much light reaches the specimen; adjusting it sharpens contrast and clarity.
Condenser	Focuses light from the source onto the specimen for even illumination.
Coarse focus knob	Raises or lowers the stage quickly to bring the specimen roughly into focus.
Fine focus knob	Moves the stage a little to sharpen the image — used for precise focusing at high power.
Arm	Supports the upper assembly and is gripped when carrying the instrument.
Base	The heavy foot that keeps the microscope steady and houses the light source.
On/off switch	Turns the built-in bulb or LED on or off.
Slide holder	Clamps the slide firmly on the stage while viewing.
Interpupillary adjuster	Sets the distance between the eyepiece tubes for comfortable binocular viewing.

■ 2.2 Magnification

Total magnification is the eyepiece magnification multiplied by the objective magnification.

Objective	Objective mag.	Eyepiece	Total
Scanning (low power)	×4	×10	×40
Low power	×10	×10	×100
High power	×40	×10	×400
Oil immersion	×100	×10	×1000

KEY POINT

A ×10 eyepiece with the ×40 objective gives a total magnification of **×400** — a favourite MCQ figure. The oil-immersion combination reaches **×1000**.

■ 2.3 Using the microscope

- » Set the microscope on the bench and remove the dust cover; connect the cable and switch on the light.
- » Adjust the interpupillary distance, then centre the condenser using the ×4 objective.
- » Place the slide on the stage and secure it with the holder.
- » Begin at the lowest power (×4). Raise the stage to its top with the coarse knob, then lower slowly until the object is in focus.
- » Step up to higher objectives as needed; at ×40 and above use **only** the fine focus knob, and set the iris diaphragm for best contrast.
- » After use: lower the stage, return to the lowest objective, switch off the light, unplug, clean the lenses with lens tissue and replace the dust cover.

■ 2.4 Care of the microscope

Category	Rule
Optical parts	Never touch a lens with the fingers. Clean only with lens tissue or soft cloth moistened with xylol.
Handling	Always carry with two hands — one on the arm, one under the base.
Storage	Keep covered and stored in a dry, stable place when not in use.
During use	Do not wet the stage with reagents, which can damage the objective.
After use	Switch off and let the bulb cool before storing; make sure slide undersides are clean before mounting.
Mechanical parts	Do not clean, lubricate or adjust mechanical parts yourself — report any damage to laboratory staff.

■ 2.5 Focusing and setting the condenser

To focus, bring the specimen into the light path and use a low-power objective ($\times 4$ or $\times 10$). Raise the stage to its uppermost point, then lower slowly until the object is sharp; move to a stronger objective to scan detail, fine-tuning with the fine focus knob. To centre the condenser on its optical axis, again use a low-power objective: close the aperture diaphragm, lower the condenser until the diaphragm opening appears in the field, bring it to the centre with the centring screws, then raise it back to its top position.

§3 PRACTICAL TECHNIQUES IN PHARMACOGNOSY

Clean microscopic preparations depend on the right clearing agents, stains and isolation methods, while micrometry allows structures to be measured rather than merely described — a key tool for telling a genuine drug from an adulterant.

■ 3.1 Clearing agents for plant pigments

Agent	Composition	Use / effect
Chloral hydrate	50 g in 20 ml water	Clarifies resistant structures; removes chlorophyll, starch, protein and mucilage while restoring collapsed tissue.
Cresol	Penetrating solvent	Penetrates prepared chalk to reveal fine structure; no separate irrigation step is needed.

■ 3.2 Chemomicroscopical reagents

Place the specimen on a slide, add one or two drops of reagent (warming if required), draw off the excess and mount in dilute glycerin.

Cell-wall stains

Stain	Result
Iodine water (with 66% v/v or 80% w/v sulphuric acid)	Stains cellulose walls blue-violet.
Phloroglucinol (1% in 90% ethanol) + HCl	Stains lignified walls red ; let the reagent evaporate before adding acid.
Tincture of Alkanna	Stains suberin (corky walls) and cutin (cuticularised walls) pink .
Corallin soda (0.25% in 25% aq. sodium carbonate)	Stains callose of phloem sieve tubes pink ; prepare fresh.

Cell-content stains

Stain	Result / use
Iodine water	Starch — blue-black ; some mucilages; proteins — yellow-brown.
Picric acid (saturated)	Proteins in dressing fibres or aleurone grains — yellow.
Tincture of Alkanna	Fixed and volatile oils — red (warm gently); fixed oils confirmed by the mount.
Ruthenium red (0.8% in 10% lead acetate)	Stains gums and mucilages red ; irrigate with 10% lead acetate.
Safranin (1% in 50% alcohol)	General tissue stain; stains nuclei and cell walls.

■ 3.3 Isolation techniques (maceration)

- » **Schultz method** (grouped lignified elements): cover a little drug with 2 ml 50% nitric acid plus a crystal of potassium chlorate, heat in a boiling water bath for 7–20 minutes, then tease apart in dilute glycerin under low power.
- » **Simple lignified elements**: add a few fragments to 2 ml 5% w/v potassium hydroxide and heat in a boiling water bath for 10 minutes, continuing until the tissue disintegrates.

■ 3.4 Micrometry

Micrometry is the measurement of objects under the microscope — for example, to separate an authentic drug from an adulterant. An **eyepiece (ocular) micrometer** is a graduated glass disc in the eyepiece, usually 10 large and 100 small divisions on an arbitrary scale. A **stage micrometer** is a slide bearing a 1 mm (1000 µm) scale split into 100 divisions of 10 µm each. Both objectives must be calibrated, and re-calibrated whenever a different graticule is used.

KEY POINT

Worked calibration. If 75 small graticule divisions align with 100 stage divisions (= 1 mm = 1000 µm), then 75 graticule divisions = 1000 µm, so **1 graticule division = 1000/75 = 13.3 µm**. Particle size is reported as a range (a–b–c) for smallest–average–largest.

■ 3.5 Preparing specimens for microscopy

Phloroglucinol/HCl staining: rinse sections in water, add a few drops of phloroglucinol then a drop of HCl, drain the excess, transfer to a drop of 50% glycerin on a clean slide and cover — such sections keep for at least a week.

Leaf-surface scraping: cut a suitable piece, lay the study surface face-down on a glazed tile, irrigate with sodium hypochlorite, hold one end with a cork, and scrape lightly at 90° with a razor blade while keeping it wet until a thin clear area is obtained; wash and brush off loose cells with a camel-hair brush.

§4

DIAGNOSTIC STRUCTURES OF PLANT MATERIAL

Whole-plant recognition rests on knowing which structures are diagnostic in each organ. This section runs from the tissues of stems and roots through the anatomy of leaves, flowers, fruits and seeds to the cell inclusions and trichomes that authenticate a drug.

■ 4.1 Parts of the living plant

A plant drug may be built from one or more of: root, rhizome (an underground stem), stem, bark, leaf, flower, fruit or seed.

■ 4.2 Roots, rhizomes and stems

In **dicotyledons (primary state)**, stems and roots carry isolated vascular bundles with defined xylem, phloem and a dividing cambium. In the **secondary state**, many dicots thicken as parenchyma between the bundles becomes meristematic, laying down rings of secondary xylem and phloem with progressive lignification of the xylem.

Monocotyledons show no secondary thickening — their vascular bundles are scattered throughout the stem.

Tissue	Characteristics
Xylem	Conducts water and mineral salts upward; lignified walls; vessels and tracheids. Vascular.
Phloem	Conducts organic nutrients up and down; sieve tubes, companion cells, phloem fibres. Vascular.
Collenchyma	Living cells with unevenly thickened cellulose walls; supports young stems. Not vascular.
Sclerenchyma	Dead, thick, lignified fibres and sclereids giving rigidity. Not vascular.
Parenchyma	Thin-walled living ground-tissue cells for storage, photosynthesis and healing. Not vascular.

NOTE

A frequent MCQ: the vascular tissues of the stem are **xylem and phloem**. Collenchyma, sclerenchyma and parenchyma provide support or storage but are **not** vascular.

■ 4.3 The leaf

Macroscopically, leaves are described by shape, size, colour, texture, odour, taste, margin, venation, surface and the presence or absence of hairs. Microscopically the diagnostic features are the waxy **cuticle**, the **epidermal cells** (often with wavy anticlinal walls), the **stomata** and their type, the **trichomes**, **calcium oxalate crystals**, and the number, size and position of the vascular veins.

Stomata type	Description	Example
Anomocytic	Surrounded by a few cells no different from the other epidermal cells (rubiaceous type).	<i>Digitalis purpurea</i> ; water-leaf
Anisocytic	Surrounded by three or four subsidiary cells, one distinctly smaller.	<i>Vigna</i> / <i>Sedum</i>
Paracytic	One or more subsidiary cells on each side, parallel to the pore's long axis.	<i>Vigna</i>
Diacytic	Two subsidiary cells whose common wall lies at right angles to the guard cells.	<i>Dianthus</i>

Two lamina arrangements are recognised: **dorsiventral** (most leaves — distinct upper and lower surfaces with palisade near the top) and **isobilateral** (palisade on both surfaces, seen in some monocots and xerophytes, e.g. senna midrib).

■ 4.4 Flowers

Commercially, “flower” can include whole inflorescences as well as single flowers. The **floral formula** uses K = calyx (sepals), C or P = petals, A = androecium (stamens) and G = gynoecium (carpels). For hibiscus: **K5, P5, A(∞), G(5)** — bisexual, five sepals, five petals (epipetalous), numerous stamens and five fused (syncarpous) carpels. Placentation may be **axile** (ovules on a central axis where septa meet), **parietal** (ovules on the wall of a one-chambered ovary) or **free-central** (ovules on a free central column).

■ 4.5 Fruits

The pericarp (fruit wall) has three layers — epicarp (outer), mesocarp (middle) and endocarp (inner). Fruits are classed as simple, aggregate or compound. Umbelliferae fruits are a **cremocarp** (a schizocarp): they split into two one-seeded mericarps, each one carpel, usually with schizogenous oil ducts (vittae) in the mesocarp.

■ 4.6 Seeds

A seed develops from a fertilised ovule and holds the embryo. The seed coat (testa plus tegmen) derives from the integuments; the **micropyle** marks the radicle's position and the **hilum** is the scar left by the seed stalk. In a bean, the parts seen in longitudinal section are the epicotyl (shoot above the cotyledon attachment), hypocotyl (below it), radicle (embryonic root), cotyledon (food-storing seed leaf) and testa.

■ 4.7 Cell inclusions

Inclusion	Description
Ca-oxalate — prismatic	Individual well-formed crystals; a very common diagnostic feature.
Ca-oxalate — raphides	Long needle-like crystals packed in parallel bundles.
Ca-oxalate — cluster (druse)	Spherical clusters of crystals, common across many families.
Starch grains	Storage polysaccharide; type, size, shape and hilum position are diagnostic (see Section 6).
Mucilage cells	Cells of gum/mucilage, common in succulents such as <i>Talinum triangulare</i> .
Fibres (lignified)	Long thick-walled sclerenchyma; stain red with phloroglucinol/HCl.
Sclereids (stone cells)	Isodiametric, heavily lignified, small-lumen, multi-pitted cells.

■ 4.8 Trichomes

Trichomes are highly variable epidermal appendages — **glandular** (secretory, with single or multicellular heads and stalks that secrete oils and resins) and **non-glandular** (covering hairs that may be single, multicellular, tufted, star-shaped or branched, sometimes with secondary walls). When shed they leave scars (cicatrices), which should not be mistaken for stomata.

§5 SCHEDULE I — FIBRES & SURGICAL DRESSINGS

Fibre identification matters because fibres go into surgical dressings, ligatures and sutures, and because adulterant fibres — even rodent or human hair — can contaminate drugs and sterile products. Official evaluation of dressings borrows the test methods used for textiles.

■ 5.1 Official quality standards

Parameters assessed for surgical dressings include adhesiveness (chiefly for plasters), area per unit weight, colour fastness, elasticity and extensibility, foreign-matter content, sterility, sulphated ash, tensile strength, thread count per stated length, fabric weight, absorbency, water-soluble extractive, water-vapour permeability and water-proofness.

■ 5.2 Chemical tests on fibres

Test	Cotton wool / gauze	Inference
4.5% NaOH, warmed to 90°C	Translucent, gelatinous appearance	Partial dissolution — confirms cellulose.
4.5% NaOH on gauze	Swollen, softer texture	Very early, incomplete dissolution.
Dilute H ₂ SO ₄ , cotton wool	No visible change	No dissolution (mild acid, room temperature).
Dilute H ₂ SO ₄ , gauze	No visible change	No dissolution.
50% w/w H ₂ SO ₄	Gradual swelling / thinning of fibres	Partial dissolution.
70% H ₂ SO ₄ , cotton wool	Thick, slightly dark gelatinous solution	Complete dissolution — confirms cellulose.
70% H ₂ SO ₄ , gauze	Clear, slightly viscous solution	Complete dissolution — confirms cellulose.

■ 5.3 Absorbency and elasticity

- » **Absorbency:** weigh 1.5 g of absorbent cotton wool, compress to about a 30 ml ball, and place it with forceps on distilled water at 20°C; record the sinking time. **BPC standard: not more than 10 seconds.** Logbook result: 8 seconds — within standard.
- » **Elasticity (crepe bandage):** clamp a 40 cm bandage so exactly 10 cm is free, stretch fully and measure, hold 1 minute, release and measure the regain. **BPC standards:** stretched length not less than twice the unstretched length; regain not more than two-thirds of the stretched length. Logbook: unstretched 10 cm, stretched 28 cm, regain 12 cm — both criteria met.

■ 5.4 Discussion

Cotton wool swells, softens and turns gelatinous with 4.5% NaOH because it is a cellulose fibre that hydrogen-bonds with alkali and takes up water; the rapid dissolution in 70% H₂SO₄ confirms cellulose through extensive acid hydrolysis. An 8-second sink time (inside the 10-second maximum) confirms high absorbency suited to wound exudate, while a stretched length of 28 cm (well above the 20 cm minimum) and a 12 cm regain (well below two-thirds of 28 cm = 18.67 cm) confirm a bandage of good, supportive elasticity.

KEY POINT

Microscopy of dressing fibres: mount in dilute glycerol with aqueous **ruthenium red** and stain with **safranin**, then record magnification. Ruthenium red targets gums/mucilage; safranin is the general stain.

§6 SCHEDULE II — STUDY OF STARCHES

Starches are polysaccharides that hydrolyse completely to glucose and fall under the general heading of carbohydrates. Hot water splits starch into an insoluble fraction (amylopectin) and a soluble fraction (amylose); the amylose is hydrolysed to maltose by β -maltase. Pharmaceutical-grade starch can be sourced from plantain (*Musa paradisiaca*), wheat (*Triticum aestivum*), breadfruit (*Artocarpus altilis*) and cassava (*Manihot utilissima*).

■ 6.1 Principal constituents

Constituent	% of starch	Characteristics
Amylose	~20%	Linear glucose chain, α -1,4 links, helical; gives the blue-black colour with iodine as the helix traps iodine.
Amylopectin	~80%	Branched — α -1,4 links with α -1,6 branch points; gives a reddish-purple / violet colour with iodine.

■ 6.2 Pharmaceutical uses

- » **Binder** — 5–10% starch paste in tablet granulation.
- » **Disintegrant** — promotes tablet breakdown in GI fluids for drug release.
- » **Diluent / filler** — inert bulking agent in tablets and capsules.
- » **Dusting powder** — on surgical gloves and to protect skin.
- » **Iodometric indicator** — turns blue-black with iodine in titrations; also an antidote in iodine poisoning.
- » **Industrial** — fabric sizing and food thickening.

■ 6.3 Microscopic granule characteristics

Feature	Maize	Potato	Wheat
Botanical name	<i>Zea mays</i> L.	<i>Solanum tuberosum</i> L.	<i>Triticum aestivum</i> L.
Family	Poaceae	Solanaceae	Poaceae
Granule shape	Polygonal / angular; compound	Large, oval to elliptical	Lenticular; bimodal (large + small)
Hilum	Central; stellate (2–5-rayed cleft)	Eccentric; cleft or branched	Central; linear crack
Lamellae	Not visible in normal light	Clearly visible (concentric rings)	Not clearly visible
Size	2–23 μ m	15–100 μ m	2–35 μ m
Polarised light	Distinct Maltese cross	Distinct Maltese cross	Distinct Maltese cross
Iodine test	Blue-black	Blue-black / blue-violet	Blue-black

KEY POINT

Granule shape is the quickest way to tell the three apart in the MCQ: **potato** = large oval with an **eccentric** hilum; **maize** = small polygonal with a central stellate hilum; **wheat** = a mixture of large and small lenticular granules.

■ 6.4 Macroscopic examination and procedure

Test	Maize	Potato	Wheat
Appearance	White, fine, silky; squeaks when pressed	White, odourless, tasteless, fine smooth	Light brown-white, smooth, slightly gritty
Cold water	Insoluble	Insoluble	Insoluble
Hot water (boil)	Thick gelatinous paste	Highly gelatinised thick paste	Viscous gel-like paste
Iodine (microscopic)	Blue-black	Blue-black / blue-violet	Blue-black

Procedure: note colour, feel, odour and taste; test cold-water solubility (~1 g in 10 ml, stir); heat to boiling for hot-water solubility and cool; for the iodine test, place grains on a slide, moisten with alcohol and dilute glycerol, cover, add dilute iodine and read the colour under the microscope.

■ 6.5 Discussion

Microscopy separates the starches cleanly by granule shape and hilum, which is exactly how adulteration is detected. All are insoluble in cold water but gelatinise on heating into viscous pastes — the property behind their use as binders and disintegrants. The blue-black iodine colour confirms amylose in every sample, so combining a simple chemical test with microscopy is an effective identification strategy.

S7

SCHEDULE III — UNORGANISED CRUDE DRUGS (GUMS)

Unorganised (acellular) crude drugs are not built into a defined plant or animal part; they are obtained by extraction or secretion — incising the opium poppy, expressing groundnut oil, or collecting beeswax or myrrh. Gums are heterogeneous protective polysaccharides that exude over plant wounds, reduce water loss, absorb water and swell, and serve as valuable formulation aids; they are identified by microscopy plus physical and chemical tests.

■ 7.1 Classes of unorganised crude drug

Category	Example	Category	Example
Dried latex	Opium	Gum-resins	Myrrh
Dried juice	Aloes	Oleo-resins	Copaiba
Extracts	Catechu	Waxes	Beeswax
Gums	Acacia	Oils & fats	Castor oil
Resins	Colophony	Volatile oils	Clove oil
Balsams	Balsam of tolu	Rubber	<i>Hevea brasiliensis</i>

■ 7.2 Acacia vs Tragacanth — the sources

Acacia gum (gum arabic) is the dried exudate from stems and branches of *Acacia senegal* (L.) Willd. and related species (Fabaceae). It is a white, odourless, bland, smooth fine powder with a light opalescent fracture, insoluble in alcohol. **Tragacanth gum** is the dried exudate of *Astragalus gummifer* Labill. and related species (Fabaceae) — pale yellowish, near-odourless, bland, rough and bumpy, appearing as white-to-yellowish flakes or ribbons.

■ 7.3 Identification tests — side by side

Property	Acacia gum	Tragacanth gum
Appearance	White, smooth powder	Pale yellow flakes/ribbons, rough
Cold-water solubility	Soluble — clear/opalescent solution	Partly soluble — swells to a gel
Hot water	Stays soluble	Forms a thick gel/mucilage
Alcohol	Insoluble	Insoluble
Flocculation with acid	Does not flocculate	Does flocculate (bassorin/tragacanthic acid)
Lead acetate	White milky precipitate	White milky, mucous precipitate
FeCl ₃ test	No visible reaction (no tannins)	Deep yellow mucilaginous precipitate
Iodine test	Blue-black only if starch contaminates	Golden-brown / yellow — starch absent
Main pharmacy use	Emulsifier, binder, stabiliser	Suspending agent, emulsifier, thickener

KEY POINT

The single fact most likely to be tested: **Tragacanth flocculates in the presence of acid** (its bassorin/tragacanthic-acid fraction), whereas Acacia does not. Both gums are insoluble in alcohol, and lead acetate gives a white precipitate with both.

§8

SCHEDULE IV — MINERAL MATTERS

Mineral matters are crude drugs of inorganic origin — acellular deposits from soil, some plain inorganic and some carrying organic components. The four studied in pharmacy are **kaolin, talc, chalk and bentonite**. Because they lack cellular structure, they are identified by organoleptic characters and physico-chemical behaviour — feel, sedimentation, crystal morphology and reaction to water — with an eye to how each is purified to pharmaceutical grade.

■ 8.1 The four minerals in brief

- » **Kaolin** (hydrated aluminium silicate) — fine white to pale-cream powder, earthy odour, greasy/plastic feel; formed by chemical weathering of feldspar-bearing rocks.
- » **Talc** (purified hydrated magnesium silicate) — very fine white-to-greyish powder, odourless and tasteless with a strongly soapy, greasy feel from its layered platelets sliding over one another.
- » **Chalk** (calcium carbonate, CaCO_3) — white, odourless, tasteless powder that effervesces with dilute HCl (releasing CO_2); of biological origin from marine shells and skeletons.
- » **Bentonite** (colloidal clay, mainly montmorillonite) — absorbs large amounts of water and swells enormously into a colloidal gel, so it will **not** sediment; this is its key distinguishing feature.

■ 8.2 Organoleptic (macroscopic) characters

Property	Kaolin	Talc	Chalk	Bentonite
Colour	White to pale cream	White to greyish-white	White	Pale cream to greyish
Odour	Faint, earthy	Odourless	Odourless	Faint, earthy
Taste	Bland, slightly gritty	Tasteless, smooth	Tasteless, slightly chalky	Bland to slightly gritty
Feel	Smooth to gritty; plastic	Very smooth, strongly soapy	Fine, slightly gritty	Smooth, fine, plastic
Palm rub	Greasy/waxy; slight slip	Slips easily; white residue	Gritty; chalky residue; no slip	Smooth, slightly plastic
Texture	Fine plastic powder	Very fine, silky (finest)	Fine, dull-white	Fine; swells to a gel

■ 8.3 Microscopic (crystal) morphology

Mineral	Mounting medium	Microscopic appearance	Mag.
Kaolin	Chloral hydrate	Irregular flat platelets, layered, greasy-looking; no distinct faces.	×40
Talc	Cresol	Thin, flat, translucent foliated/lamellar platelets; overlapping layers.	×40
Chalk	Chloral hydrate	Irregular rounded to angular white granules; no distinct crystal faces.	×40
Bentonite	Chloral hydrate	Extremely small platy particles in a gel-like matrix; hard to resolve.	×40

■ 8.4 Sedimentation rate

Weigh 1.0 g of mineral, transfer to a 200 ml measuring cylinder, make up to the mark with distilled water, stopper and shake, then time how long the solid takes to settle completely. The rate is calculated as:

KEY POINT

Sedimentation rate (g/s) = mass of specimen (g) ÷ sedimentation time (seconds). Example: 1.0 g of kaolin settling in 100 s gives SR = 1.0/100 = 0.01 g/s. Bentonite has **no** calculable rate — it forms a stable colloidal gel and does not settle.

Mineral	Mass	Time	Rate (g/s)	Observation
Kaolin	1.0 g	~60–120 s	~0.008–0.017	Settles gradually into a compact white layer.
Talc	1.0 g	~120–180 s	~0.006–0.008	Settles slowly; very fine platelets float briefly.
Chalk	1.0 g	~30–60 s	~0.017–0.033	Settles quickest — denser granular particles.
Bentonite	1.0 g	Did not sediment	Not applicable	Swells to a colloidal gel; stays suspended.

Sedimentation rate is governed by **particle size** (smaller settle slower, per Stokes' law), **density** (denser settle faster) and **particle shape** (irregular shapes settle slower). It is **not** determined by the mineral's source or chemical composition — so the statement "determined by source/composition" is FALSE. Solubility does not affect it either, since these minerals are practically insoluble.

■ 8.5 Pharmaceutical applications

Mineral	Uses	Example products
Kaolin	Adsorbent antidiarrhoeal (adsorbs toxins/bacteria); tablet diluent; poultice base; emollient in dusting powders.	Kaolin & Morphine Mixture BP; Kaolin Mixture BP.
Talc	Glidant/lubricant in tablets; dusting-powder base; inert excipient; filtration aid.	Purified Talc BP; Compound Dusting Powder BP.
Chalk	Antacid (neutralises gastric HCl); calcium supplement; toothpaste abrasive; tablet diluent.	Chalk Mixture BP; Compound Chalk Powder BP.
Bentonite	Suspending agent for insoluble powders; skin mask/poultice (antibacterial); thickener; colloidal adsorbent.	Calamine Lotion BP (suspending agent); Bentonite Magma BP.

■ 8.6 Precautions

- » Weigh specimens on a calibrated balance for reliable sedimentation calculations, and shake cylinders thoroughly before timing.
- » Handle cresol carefully — it is a skin irritant; place cover slips at an angle to avoid air bubbles.
- » Clean lenses only with lens tissue; taste only a trace on the tongue tip and never swallow; handle each mineral in a separate bench area to avoid cross-contamination.

§9

SCHEDULE V — EXAMINATION OF CRUDE DRUGS

Crude drugs are minimally processed natural substances, and their examination establishes authenticity, purity, quality and fitness for medicinal use — guarding against adulteration or substitution. Each specimen is described by common name, botanical name (genus, species, author), family, morphological part, form, major chemical constituents and a simple sketch.

■ 9.1 Beans

Parameter	Details
Common / botanical name	Bean; <i>Phaseolus vulgaris</i> L.
Family	Fabaceae (Leguminosae) — legume/pea family
Morphological part	Seed coat, two cotyledons, embryo (hypocotyl + plumule) and a seed scar (hilum).
Constituents	15–30% protein; 60–70% carbohydrate; 0.7–2% lipid.
Commercially available	India, Brazil and Myanmar.

■ 9.2 Turmeric

Parameter	Details
Common / botanical name	Turmeric, yellow root; <i>Curcuma longa</i> L. (Yoruba: <i>atele pupo</i>)
Family	Zingiberaceae (ginger family)
Morphological part	The rhizome — a thick, branching underground stem used as a spice.
Constituents	Curcuminoids (curcumin, demethoxycurcumin, bisdemethoxycurcumin) plus volatile oils.
Relevance	Curcumin is anti-inflammatory, antioxidant and antimicrobial; volatile oils add aroma.

Macroscopy: cylindrical, irregular, often branched rhizome segments; orange-brown outer skin; rough, dry, wrinkled surface with visible nodes and ridges; mild characteristic aroma. **Internally** (after longitudinal section): deep orange, slightly moist, giving a yellow-orange juice when pressed.

■ 9.3 Hibiscus flower

Parameter	Details
Common / botanical name	Roselle, red tea, Jamaican sorrel, karkadi; <i>Hibiscus rosa-sinensis</i> L. (Nigeria: <i>zobo</i>)
Family	Malvaceae
Placentation	Axile
Floral formula	K5, P5, A(∞), G(5) — bisexual; 5 sepals, 5 petals at base, numerous stamens, 5 fused carpels.
Constituents	Flavonoids, anthocyanins, alkaloids, amino acids, vitamin C.

■ 9.4 Fruits — pawpaw and okra

Parameter	Pawpaw	Okra
Botanical name	<i>Carica papaya</i> L.	<i>Abelmoschus esculentus</i>
Common / vernacular	Pawpaw / <i>ibepe</i>	Okra, lady's finger / <i>ila</i>
Family	Caricaceae	Malvaceae
Description	Large, oval-to-oblong, smooth green/yellow surface.	Elongated, green, ridged, tapering.
Placentation	Axile / parietal	Axile
Seeds	Small, black, round, coated with a mucilaginous layer.	Round, soft; release sticky mucilage when pressed.

■ 9.5 Leaves — false ashoka

Parameter	Details
Botanical name	<i>Polyalthia longifolia</i> (Sonn.) Thwaites
Common name	False ashoka, masquerade tree, mast tree, Indian fir tree
Family	Annonaceae
Leaf characters	Glossy green, wavy margins, smooth leathery texture, lanceolate; alternate phyllotaxy.
Constituents	Alkaloids, tannins, flavonoids and saponins.

§10 SCHEDULE VI — MACROSCOPY OF CRUDE VEGETABLE DRUGS

Crude vegetable drugs are major sources of active ingredients — tea, coffee and kolanut, for instance, all supply caffeine, an ingredient in analgesics such as Pancemol, Polfanol, APC tablets and Cafergot. Macroscopic evaluation is among the most reliable identification methods, giving preliminary information before microscopic, chemical or physical tests.

■ 10.1 Diagnostic (organoleptic) characters

Character	Description
Morphological	Shape, size, fracture, surface, margin, venation, arrangement.
Colour	Observed under natural light, before and after cutting.
Odour	Characteristic smell, released by breaking, crushing or bruising.
Taste	Non-toxic drugs only — bland, bitter, sweet, sour, astringent or acid.
Texture	Smooth, rough, hairy, gritty, powdery, fibrous, waxy or greasy.
Exudate	Latex, gum, resin or oil released on cutting or pressing.

■ 10.2 Leaf-drug diagnostic characters

- » **Venation:** pinnate (one midrib with laterals), palmate (veins radiating from the base) or parallel (monocot).
- » **Surface:** upper and lower epidermis, cuticle, and the presence and type of trichomes.
- » **Stomata:** type (anomocytic, anisocytic, paracytic, diacytic) and distribution across the surfaces.
- » **Margin:** entire, serrated, dentate or lobate. **Arrangement:** alternate, opposite or whorled.

■ 10.3 *Tithonia diversifolia* (Mexican sunflower) — stem bark

- » **Calcium oxalate:** prismatic crystals, very obvious in transverse section.
- » **Lignified fibres:** long, thick-walled, very narrow lumen; stain deep red with phloroglucinol; bundled within the phloem.
- » **Scleroids:** very small lumen, isodiametric, thick-walled, with pitted walls.
- » **Vascular bundles:** a distinct ring.

■ 10.4 *Talinum triangulare* (water leaf)

Water leaf (family Talinaceae) is a succulent. Both leaf surfaces bear thin-walled polygonal cells with **anomocytic** stomata — more abundant on the lower epidermis — and trichomes are generally absent; the upper surface shows very obvious raphide crystals of calcium oxalate. In the stem bark, raphides are common (parallel packed needles), fibres are thin to moderately thick-walled and stain red with safranin (less lignified than in woody species), scleroids are few, and the cortex is thin-walled, horizontally elliptical, compact parenchyma.

■ 10.5 Macroscopy of fruits — exercise

- » Describe external features — shape, colour, size, texture, odour and taste.

- » Break the fruit and note the splitting pattern (suture lines, dehiscence type).
- » Cut longitudinally to expose placentation, seed arrangement and any exudate, then draw and label the L.S.

§11 STUDENT EXPERIMENT RESULTS SUMMARY

A run-through of the seven practicals completed during the course, with the aim, materials/method and the conclusion drawn from each.

Exp. 1 (6 Oct 2025) — Use & Care of the Microscope

Aim: to learn to use and care for the microscope. The practical gave a clear grasp of its parts, functions, uses and safe handling — essential groundwork for all later experiments.

Exp. 2 (20 Oct 2025) — Examination of Crude Drugs

Using beans, turmeric, CARTEF-DS 30/400, Planting 45 powder, Senokot, Penicillin ointment and Passion talcum powder, the colour, shape and size of each proprietary product were examined and their botanical/common names, active constituents and natural sources identified. Conclusion: macroscopic evaluation is a fundamental skill giving essential preliminary information for identification, authentication and quality assessment.

Exp. 3 (3 Nov 2025) — Macroscopy of Crude Vegetable Drugs

Aim: to identify crude vegetable drugs by their macroscopic characters. A hibiscus flower was sectioned longitudinally with a razor and its floral parts observed under a hand lens. Conclusion: clear macroscopic characters (shape, size, colour, texture, arrangement) were identified.

Exp. 4 (17 Nov 2025) — Diagnostic Characters of Medicinal Plants

Tithonia diversifolia and *Talinum triangulare* were examined and their transverse sections stained with safranin. *Tithonia* showed rough, serrated leaves with abundant trichomes, a thick cuticle, anomocytic stomata and well-developed vascular bundles — a hardy, dry-adapted plant. *Talinum* showed the smooth, soft, entire-margined leaves of a succulent, with thin-walled epidermal cells, mucilage cells and visible raphides.

Exp. 5 (16 Mar 2026) — Unorganised Crude Drugs (Acacia & Tragacanth)

General, solubility, flocculation and identification tests gave results consistent with the known characters of Acacia gum (*Acacia senegal*) and Tragacanth gum (*Astragalus gummifer*) as unorganised crude drugs.

Exp. 6 (30 Mar 2026) — Study of Starches

Maize, potato and wheat starches were identified by macroscopic, microscopic and iodine methods. Potato was recognised by large oval granules with an eccentric hilum, maize by small polygonal granules, wheat by mixed granule sizes; the iodine test gave blue-black in all, confirming starch.

Exp. 7 (13 Apr 2026) — Fibres & Surgical Dressings

The physical and chemical properties of dressing fibres were evaluated. Cotton wool (absorbency) and gauze (support/bandaging) each proved well suited to surgical dressing and wound care.

§12 QUICK-RECALL FLASH FACTS FOR THE MCQ

Condensed question-and-answer tables covering the whole course — ideal for last-minute revision before the MCQ test.

12.1 Microscope

Question	Answer
Magnification of the ocular lens	×10
Total magnification with the ×40 objective	×400
Part controlling the amount of light	Iris diaphragm
Suitable stain for internal cell identification	Safranin
Mounting medium for microscopy	Glycerol / dilute glycerin
“Glycerol is added before mounting” — T/F	TRUE — for proper irrigation of the sample

12.2 Mineral matters

Question	Answer
Mineral that does NOT sediment	Bentonite (swells to a colloidal gel)
Sedimentation rate is determined by	Particle size, density and shape
“Rate depends on source/composition” — T/F	FALSE
Factor NOT affecting the rate	Solubility (minerals are practically insoluble)
Mineral from chemical weathering of rocks	Kaolin
Sedimentation-rate formula	SR = mass (g) ÷ time (s)
Mineral with a soapy/greasy feel	Talc
Mineral that effervesces with dilute HCl	Chalk (CaCO ₃)
Bentonite is used as	Suspending agent; skin mask/poultice
Kaolin is used as	Adsorbent antidiarrhoeal; tablet diluent

12.3 Organised vs unorganised drugs

Question	Answer
Unorganised crude drugs are also called	Acellular drugs
Part with stomata, venation, chlorophyll, cuticle	Leaf
Characters examined by the sense organs	Macroscopic (organoleptic) characters
Stomatal number/type are which characters	Pharmacognostic (diagnostic) / histological
Vascular tissues of the stem	Xylem and phloem
Unorganised drugs include all EXCEPT	Senokot tablet (a manufactured product)
Tragacanth flocculates in the presence of	Acid (bassorin / tragacanthic acid)

12.4 Starch

Question	Answer
Starch is approximately	80% amylopectin, 20% amylose
Component giving blue-black with iodine	Amylose (linear; traps iodine in its helix)
Amylopectin colour with iodine	Reddish-purple / violet
Fehling's reagent tests for	Reducing sugars (brick-red Cu_2O precipitate)
Ferric chloride detects	Phenolic compounds (blue/green/purple)
Maize starch	Polygonal; central stellate hilum; 2–23 μm
Potato starch	Large oval; eccentric hilum; 15–100 μm
Wheat starch	Lenticular, bimodal; central linear hilum
Botanical names	Plantain <i>Musa paradisiaca</i> ; potato <i>Solanum tuberosum</i> ; wheat <i>Triticum aestivum</i> ; maize <i>Zea mays</i>

12.5 Surgical dressings

Question	Answer
BPC absorbency standard	Cotton sinks in not more than 10 s
Elastic bandage — stretched length	Not less than twice the unstretched length
Elastic bandage — regain	Not more than 2/3 of the stretched length
70% H_2SO_4 on cotton fibres	Complete dissolution — confirms cellulose
4.5% NaOH (warm) on cotton	Translucent/gelatinous — confirms cellulose
Stain for fibre microscopy	Ruthenium red (gums/mucilage); safranin (general)

12.6 Gums (Acacia vs Tragacanth)

Question	Answer
Acacia in cold water	Soluble — clear/opalescent solution
Tragacanth in cold water	Partly soluble — swells, does not fully dissolve
Both gums in alcohol	Insoluble
Gum that flocculates with acid	Tragacanth
FeCl_3 on Acacia	No visible reaction (no tannins)
FeCl_3 on Tragacanth	Deep yellow mucilaginous precipitate
Lead acetate on both	White milky precipitate — confirms gum
Three properties to identify unorganised drugs	Colour, odour, solubility

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